

Background Information

Symptoms [1]

Major Motor Symptoms:

- Bradykinesia , Rigidity , Tremor , Gait alterations, Posture alterations, Freezing of gait, Balance alterations

Major non-motor symptoms:

- Hyposmia, Sleep disorders, Neuropsychiatric features, Autonomic dysfunction, Mild cognitive impairment, Pain and somatosensory disturbances, Dementia

PD underlying mechanism

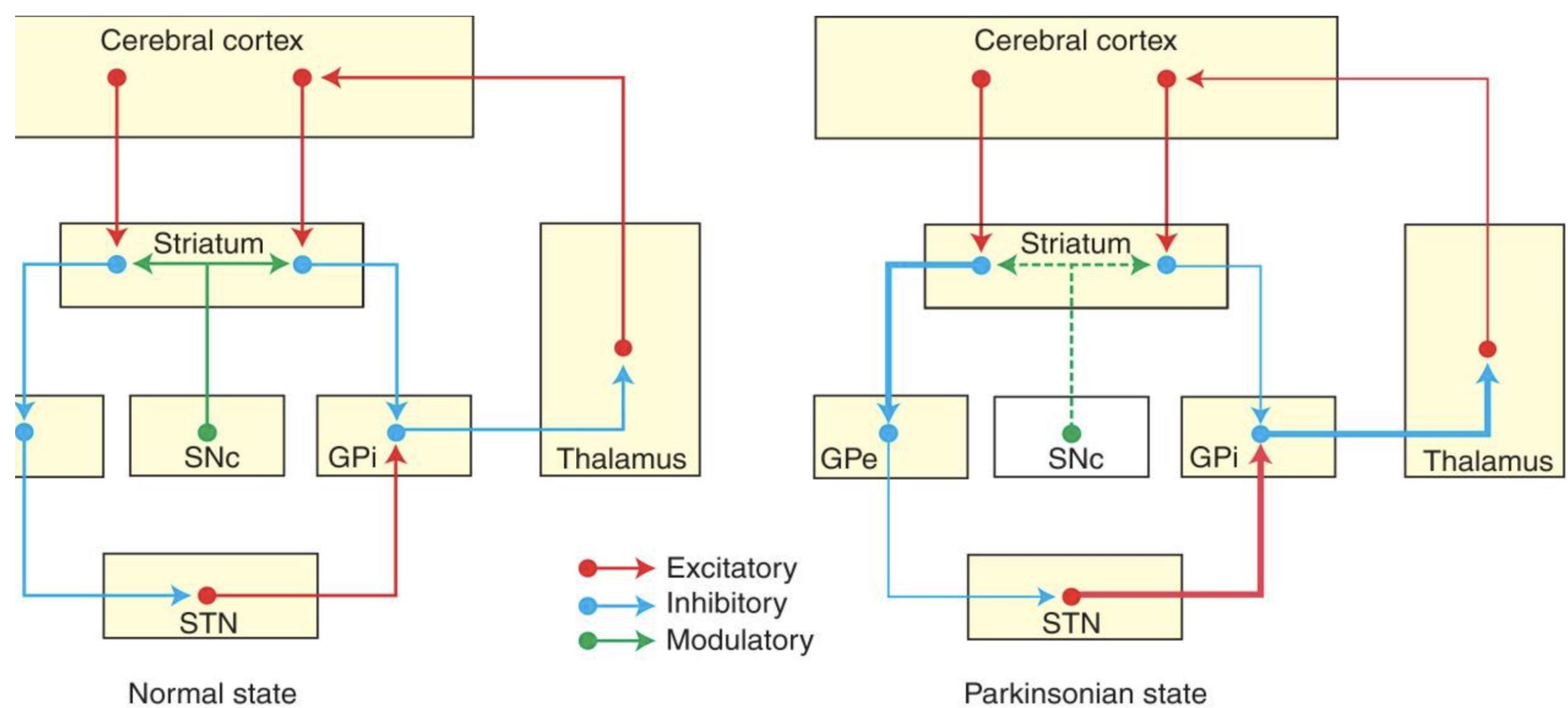


Figure 1. PD Mechanism Dopamine Neuron Deficiency [2]

Existing Solutions

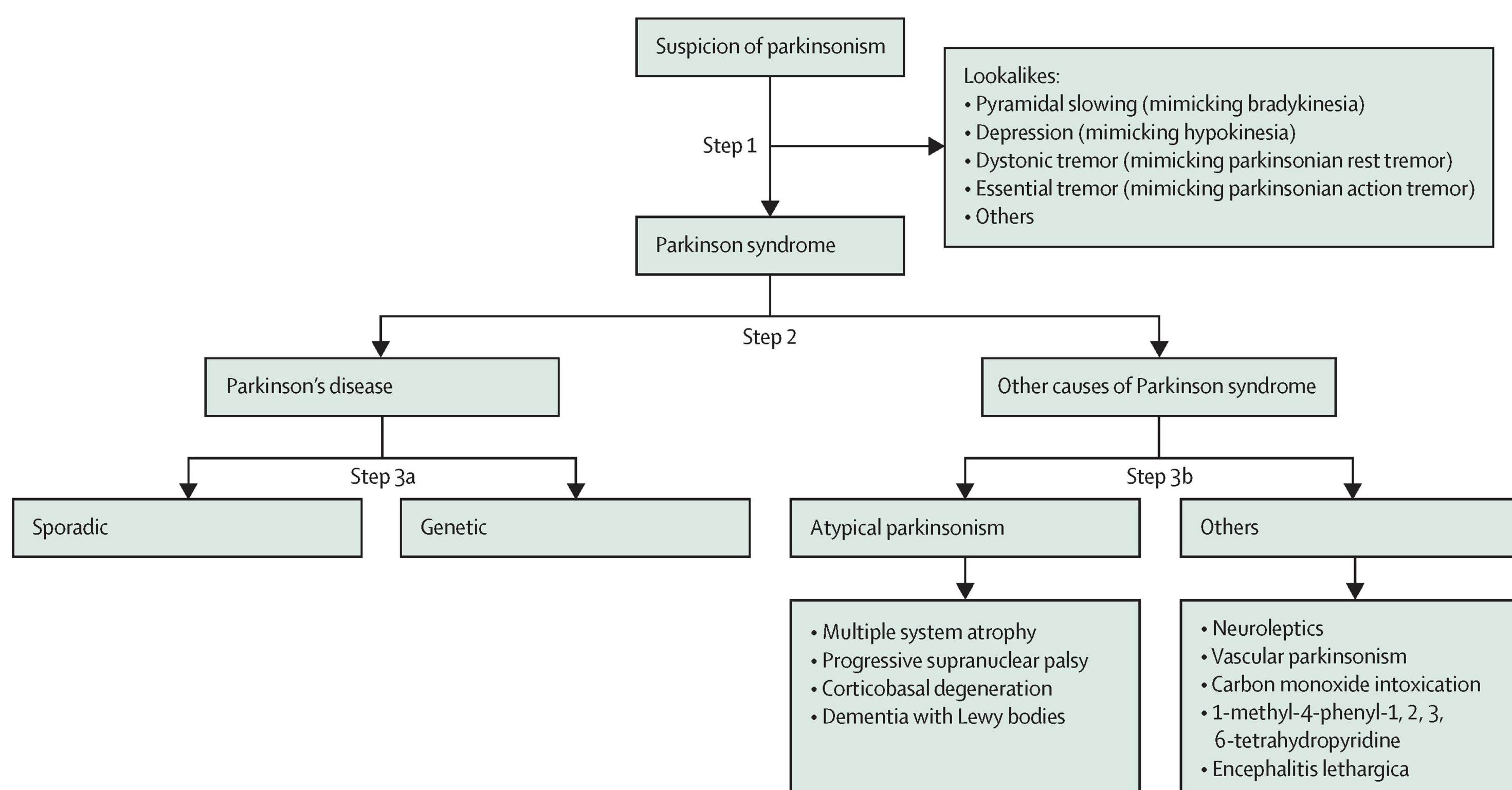


Figure 2. Clinical Diagnosis Process of Parkinson's Disease [3]

Problems with existing solutions [1]

- Accurate diagnosis usually happens at late stage
- Early-stage diagnosis is inaccurate
- Neither of current solutions allows consecutive measurement

Proposed Solution

Accelerometer wristband assisting early detection of Parkinson's Disease

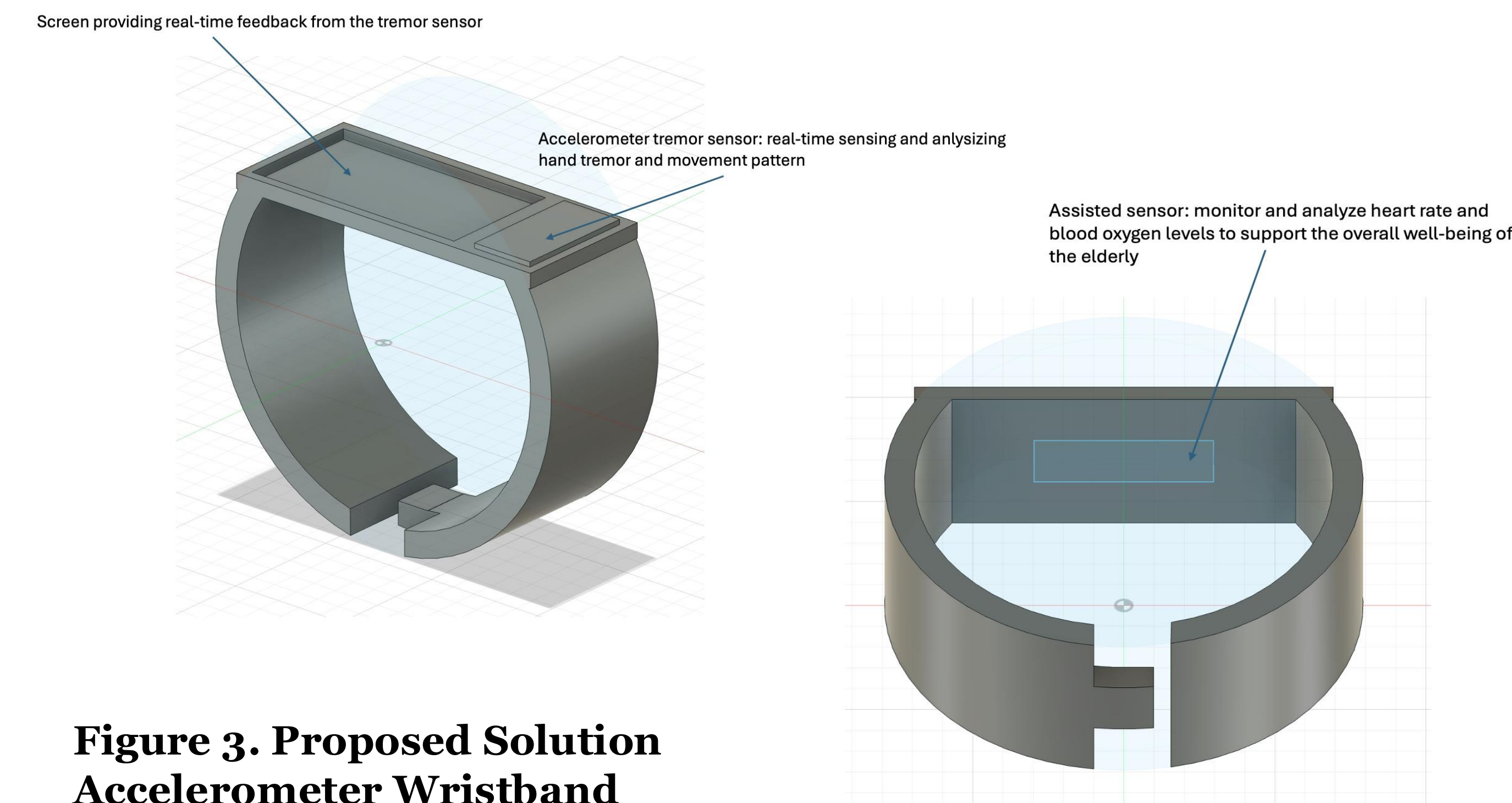


Figure 3. Proposed Solution Accelerometer Wristband

- Accelerometer tremor sensor: record acceleration of hand and wrist movements in real time
- Assisted sensor: monitor daily vital signs such as blood oxygen levels and heart rate, enhancing the functionality and practicality of the wristband
- Processor chip: analyze recorded data from sensors; output results through the screen
- Screen: provide real-time feedback on data received from the sensors, facilitating efficient human-machine interaction

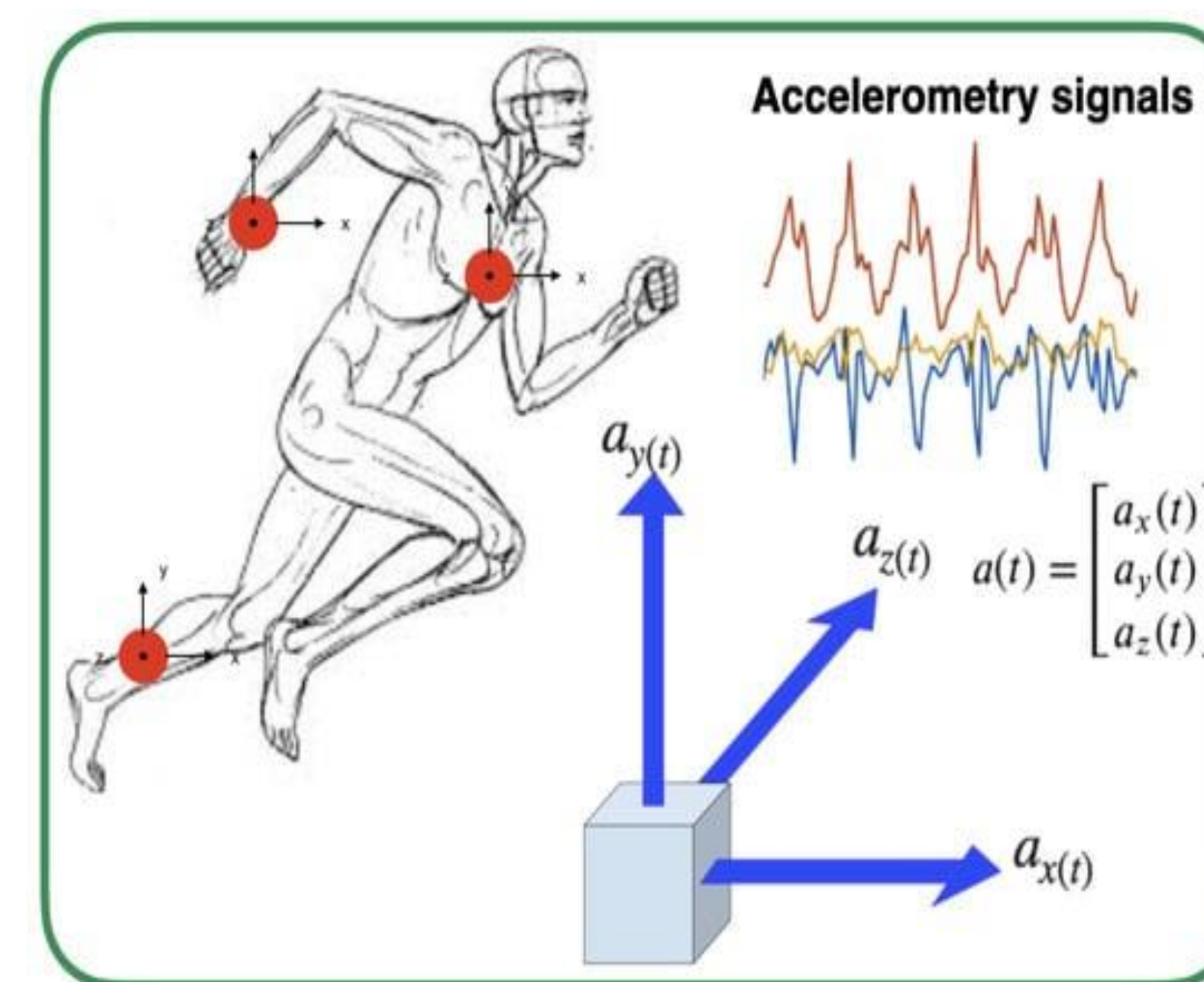


Figure 4. Accelerometers Detect Accelerations of Human Movements [4]

Mechanism

- Rest tremor and low amplitude hand action tremor occur at early stages of PD [1]
- Reduction in wrist acceleration observed years before clinical diagnosis of PD [5]
- Such reduction is unique to PD, and predicts prodromal PD better than other markers [5]
- Accelerometers can detect acceleration in axes of space, enabling modeling of hand movements [4]

Market Analysis

Large potentially affected population

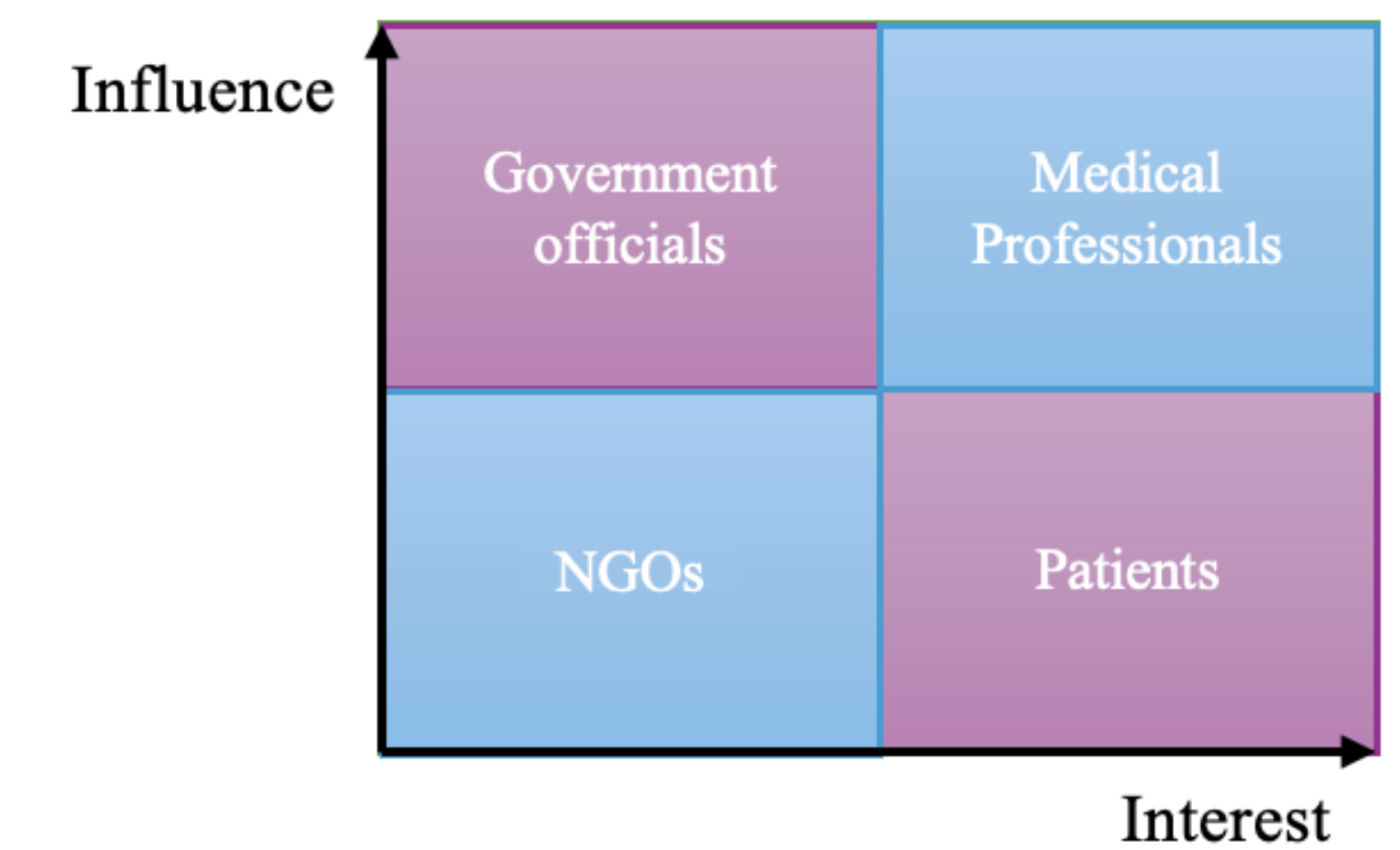


Figure 5. Stakeholder Map Showing Relative Interest and Influence for Different Related Groups

Necessity of early detection of PD

- Greatest chance of success for disease-modifying therapies during early stage [1]
- Prevent significant decrease in patients' quality of life [1]
- Enable identification of participants for future clinical research [5]

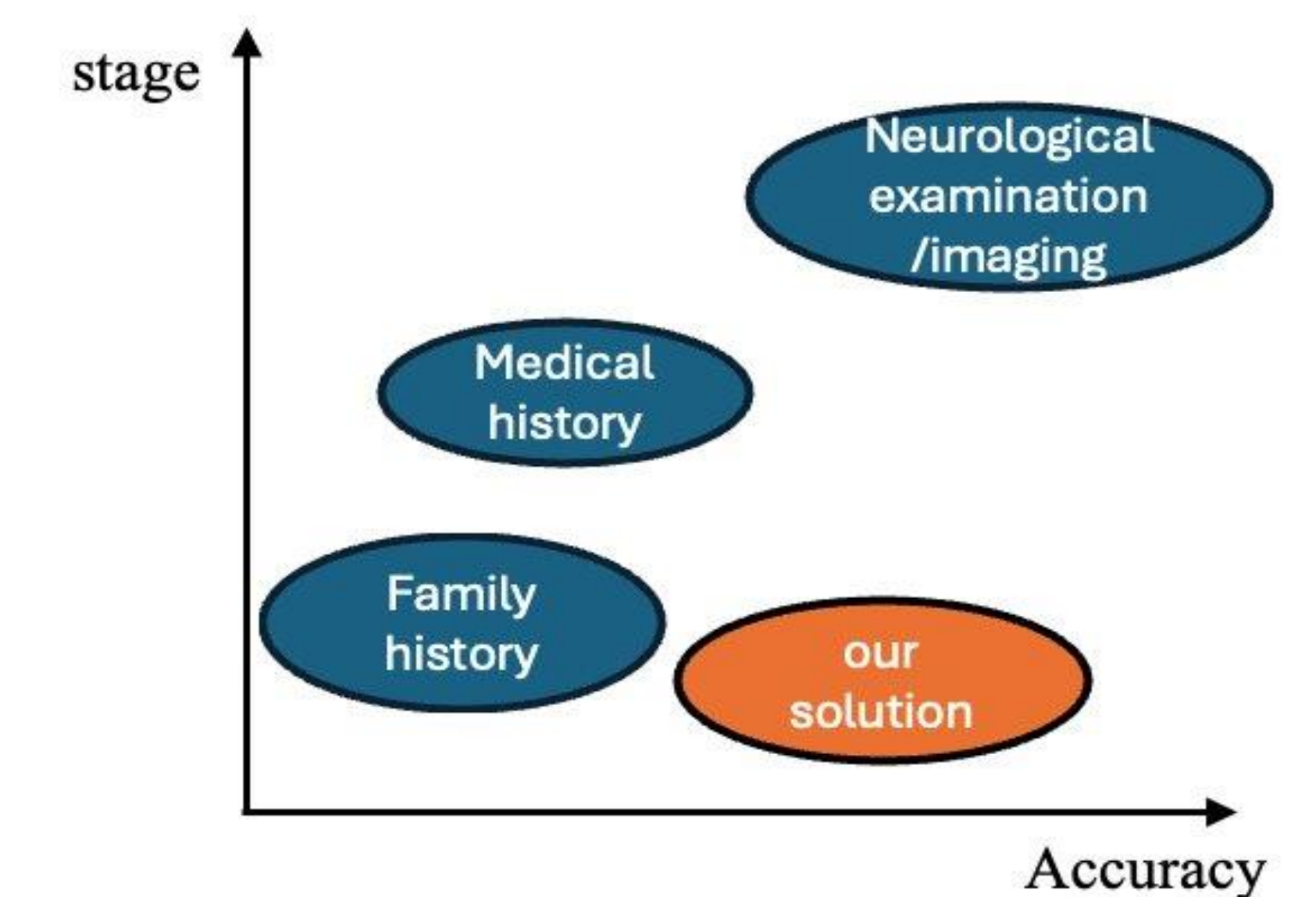


Figure 6. Ways to diagnose Parkinson's disease vs accuracy

Reference

[1]Tolosa, Eduardo et al. "Challenges in the diagnosis of Parkinson's disease." *The Lancet. Neurology* vol. 20,5 (2021): 385-397. doi:10.1016/S1474-4422(21)00030-2

[2] Lanciego, José L et al. "Functional neuroanatomy of the basal ganglia." *Cold Spring Harbor perspectives in medicine* vol. 2,12 a009621. 1 Dec. 2012, doi:10.1101/cshperspect.a009621

[3]Bloem, B. R., Okun, M. S., & Klein, C. (2021). "Parkinson's disease". *The Lancet*, 397(10291), 2284–2303. [https://doi.org/10.1016/S0140-6736\(21\)00218-X](https://doi.org/10.1016/S0140-6736(21)00218-X)

[4]Cavita-Huerta, E., Reyes-Reyes, J., Romero-Ugalde, H. M., Osorio-Gordillo, G. L., Escobar-Jiménez, R. F., & Alvarado-Martínez, V. M. (2024). "Human Activity Recognition from Accelerometry, Based on a Radius of Curvature Feature." *Mathematical and Computational Applications*, 29(5), 80. <https://doi.org/10.3390/mca29050080>

[5]Schalkamp, AK., Peall, K.J., Harrison, N.A. et al. "Wearable movement-tracking data identify Parkinson's disease years before clinical diagnosis." *Nat Med* 29, 2048–2056 (2023). <https://doi.org/10.1038/s41591-023-02440-2>